Introduction:

The Dencun Update:

The Dencun update, is a significant milestone in the Ethereum 2.0 roadmap. It introduces several key features to the Ethereum network:

1. **Sharding**: This long-awaited scaling solution will help Ethereum achieve higher throughput by partitioning the network into multiple shards or parallel blockchains. Transactions will be processed across shards, increasing overall capacity.

2. **Verkle Trees**: A novel data structure that enables efficient encoding and transfer of Ethereum's state across shards. Verkle Trees **optimize storage and bandwidth requirements**.

3. **Protodank Transactions**: A new transaction format designed to be more efficient and compact, reducing blockchain bloat and gas fees. Protodank Transactions leverage **advanced cryptography**.

4. **Blob Transactions**: Large contract deployments and data storage will be offloaded to a separate "blob" layer, **freeing up space on the main execution layer**.

5. **Stateless Clients**: Node operators will no longer need to store the entire Ethereum state, significantly **reducing hardware requirements for running nodes**.

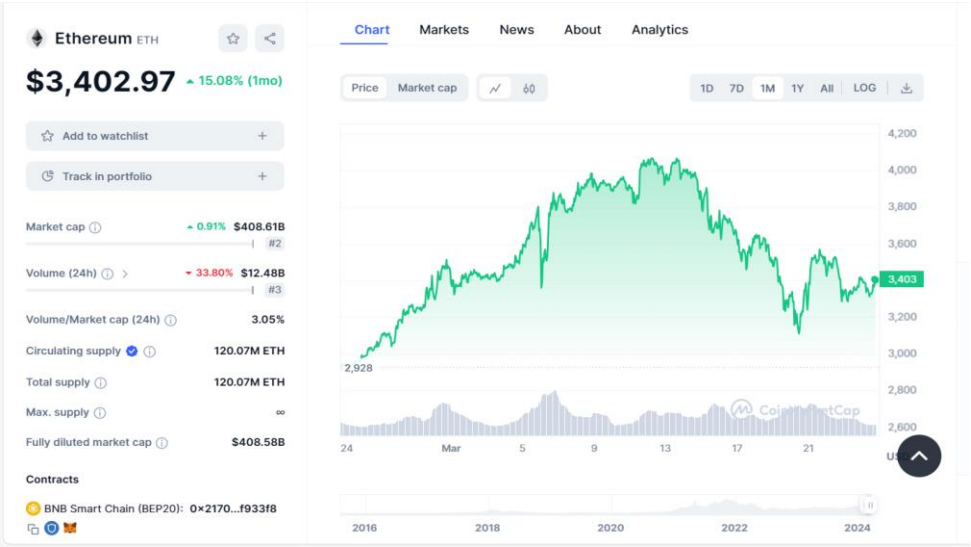
Extract - The Dencun update aims to make Ethereum more **scalable, efficient, and decentralized**, paving the way for broader adoption and more complex decentralized applications.

Immediate effects cause by the Dencun update:

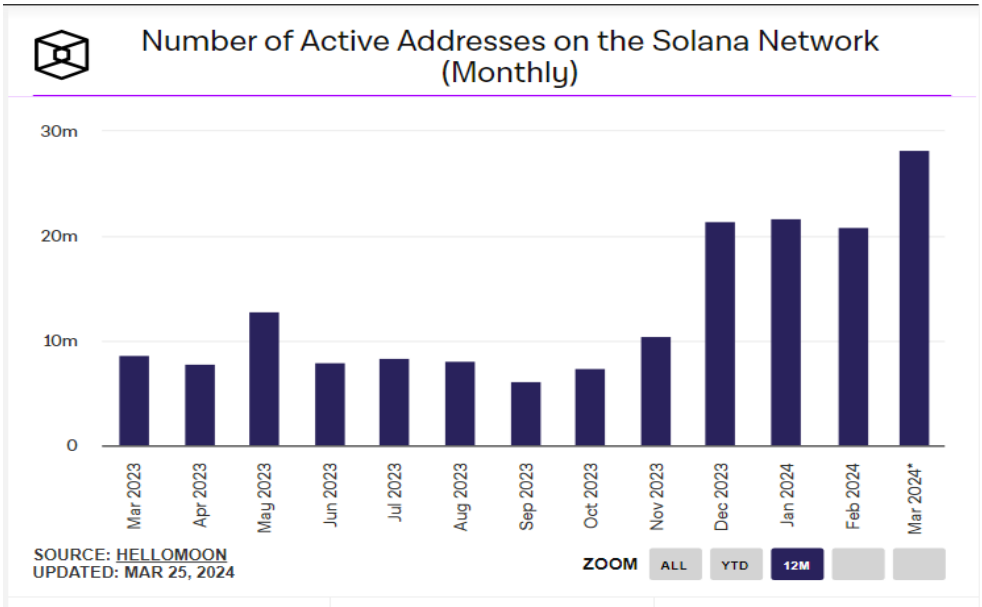
Ethereum's price **plunged** after the Dencun update. Analysts debate the reasons:

- **Unmet expectations**
- **Technical glitches**
- **Profit-taking**
- **Wider market woes**

The update might have triggered a domino effect on other coins. Only time will tell if this is a temporary dip or a long-term reversal. Let's find out...



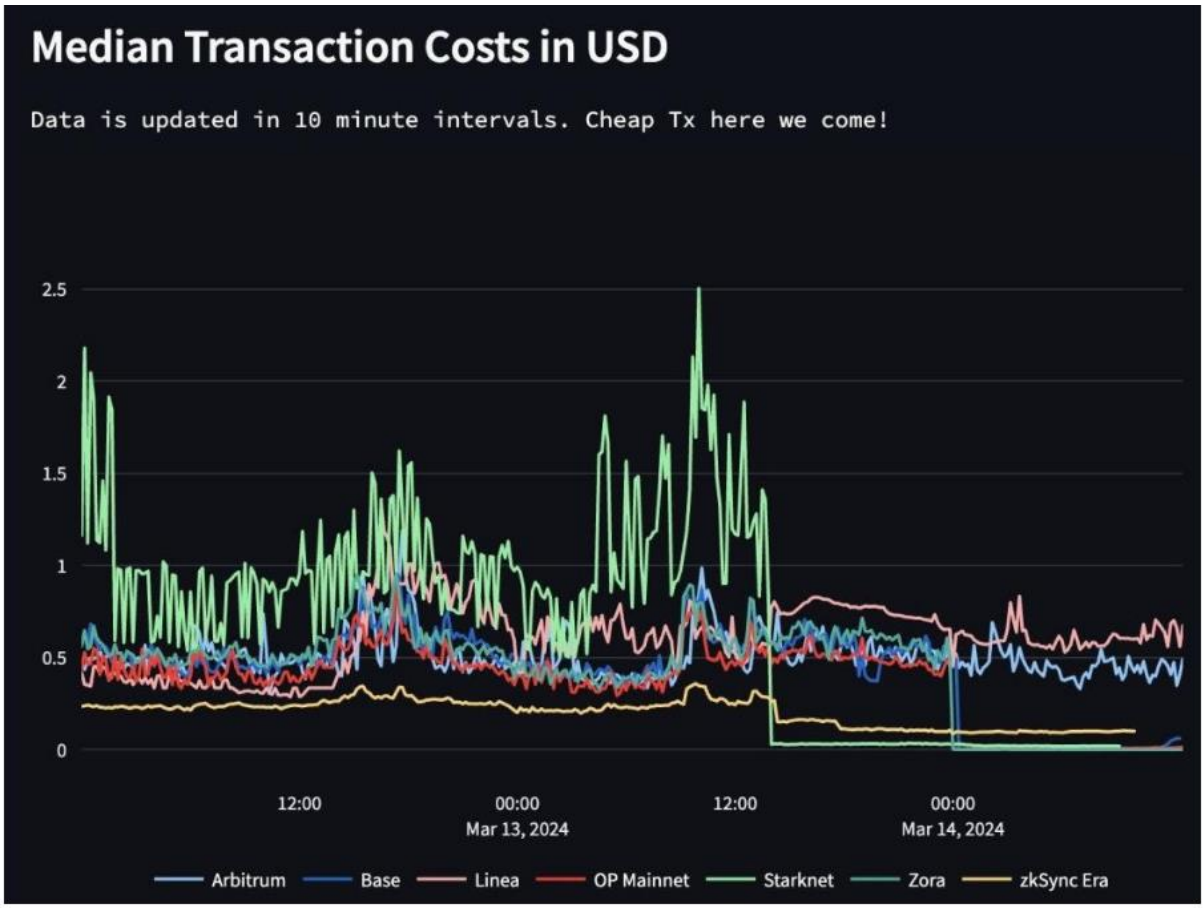
Immediate Effect - The Ethereum markets sank following the dencun update. Reason: Unmet expectations and mitigation to Solana by investors.



Immediate Effect - Rise in Solana's active addresses which cites to increase in use of Solana's layer 1 blockchain.

Technical Insights:

- **Proto-danksharding (EIP 4844)**: The main innovation in Dencun is **proto-danksharding**, a scaling solution that aims to **reduce transaction fees on Ethereum rollups by up to 100 times**. Unlike the earlier ETH2 sharding plan, which split Ethereum into 64 shards, proto-danksharding focuses on rollups. It introduces **blobs**, large objects filled with transaction data that can't be read by the Ethereum Virtual Machine (EVM). Blobs eliminate the need to write transaction data in the form of calldata, which used to account for most of the rollup costs. As a result, transaction fees on rollups have dropped significantly after the upgrade.



Graph representing the change in transaction cost and fees of Ethereum layer 2 tokens.

- **24-Hour Blob Production**: Within 24 hours of the upgrade, there were an average of **1.39 blobs sent to Ethereum every hour**, compared to the target of 3 blobs/hour. Starknet accounted for almost 77% of all blobs.

- **Ethereum Mainnet Fees**: It’s important to note that Ethereum mainnet fees didn’t go down after the upgrade; only rollup fees will. However, Pontem’s new L2, **SuperLumio**, should also benefit from proto-danksharding. While the price of ETH did revisit $4,000 just before the mainnet release. In the longer term, if blob production reaches the target and fees remain low, Cancun will bring benefits to the entire rollup ecosystem.

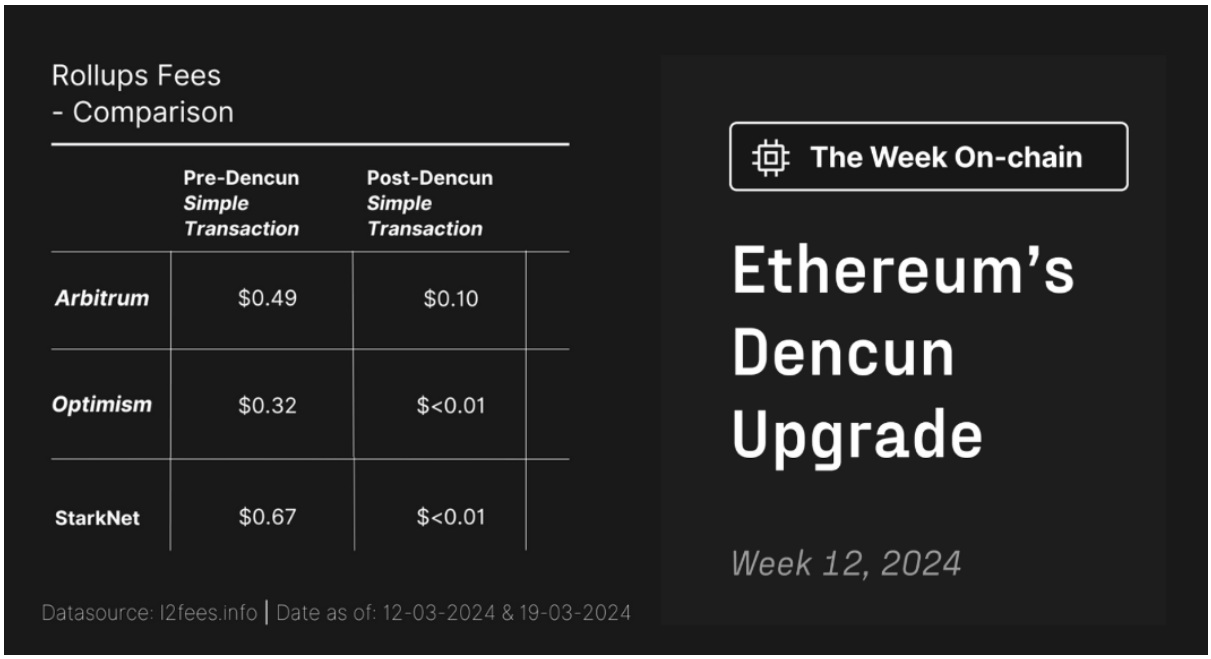


Image representing the change in per transaction cost of layer-2 coins of Ethereum network.

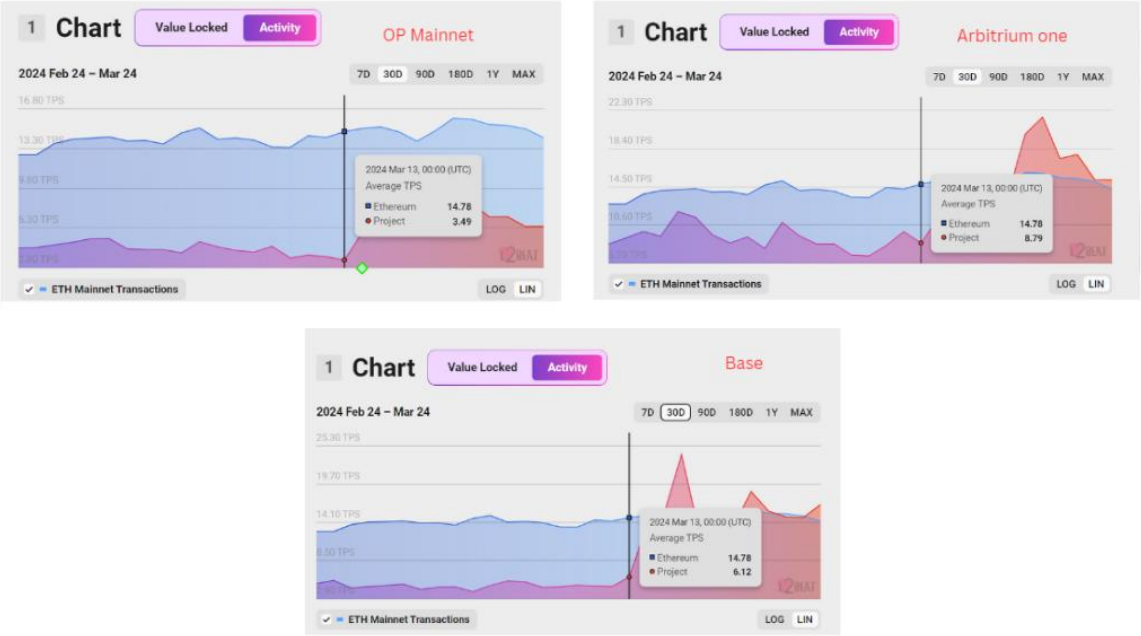
Conclusion:

The Dencun update for Ethereum aimed to address scalability concerns and pave the way for a more robust future. While the update has demonstrably improved the situation, it's crucial to acknowledge the emergence of a potential long-term trend within the Ethereum ecosystem.

Layer 2 Solutions and Alternative Chains: A Breeding Ground for Innovation

The Dencun update, despite its improvements, hasn't entirely eliminated the limitations of the Ethereum main chain. Transaction fees of Layer 1 is still a barrier for some users. This creates fertile ground for Layer 2 scaling solutions and alternative blockchains with faster transaction speeds and lower fees. We can expect continued growth and innovation in these areas, attracting new developers and

investors seeking a more user-friendly environment for building decentralized applications (dApps). Chains like Solana, with their inherent scalability features, might also benefit from this shift.



*Images above represent the Exponential rise in total transaction counts of the layer-2 of Ethereum. This shows **layer-2 transactions exceeds layer-1s** after the update.*

Ethereum Main Chain: A Store of Value and Settlement Layer

The Dencun update, coupled with future upgrades like Sharding, could solidify Ethereum's position as a store of value similar to Bitcoin and gold. Its established network effect, security, and large user base make it a prime candidate for holding digital assets over the long term. Additionally, the main chain might evolve into a settlement layer for high-value transactions, leveraging its security advantages even with lower transaction frequency.

A Bifurcated Future: Specialization and Synergy

The Dencun update marks a significant chapter in Ethereum's evolution, but it also highlights a potential bifurcation within the ecosystem. Layer 2 solutions and alternative chains might cater to the needs of fast-paced, high-volume applications, while the Ethereum main chain transforms into a secure settlement layer and store of value. This doesn't necessarily signify competition; instead, it could foster a symbiotic relationship where Layer 2 solutions and other chains leverage the security of the Ethereum main chain, while the main chain benefits from the innovation

happening at the Layer 2 and alternative chain levels. Ultimately, this specialization could lead to a more robust and scalable blockchain ecosystem overall.

Call:

Ethereum Main Chain - Mildly Bearish in long-term

Reason: Layer-2 scaling solutions and other blockchains like Solana are offering faster and cheaper transactions, potentially attracting users away from the Ethereum main chain in the long run.

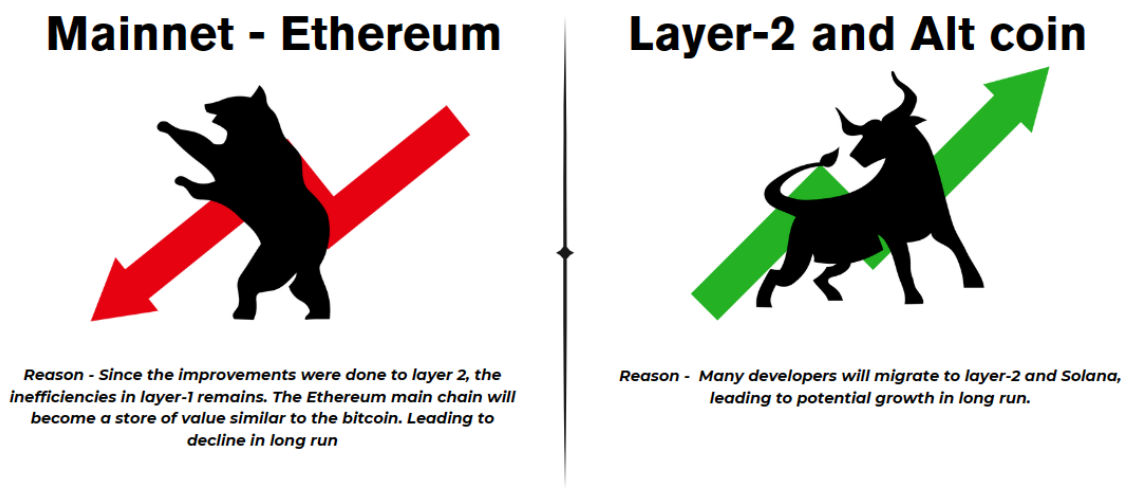
Ethereum layer-2 tokens -Bullish in long-term

Reason: Layer-2 solutions like Polygon, Optimism, and Arbitrum are built on top of Ethereum, inheriting its security while offering significant improvements in transaction speed and fees. The Dencun update could further strengthen this integration.

Ethereum's existing DeFi (Decentralized Finance) and NFT (Non-Fungible Token) ecosystem is vast and established. Layer-2 solutions can make these applications more accessible and scalable, driving growth in the long term.

Solana - Bullish in long-term

Reason: Solana offers significantly faster transaction speeds and lower fees compared to Ethereum's main chain, making it attractive for developers and users seeking a more efficient platform.



The short-term market movements are dependent on many factors. Since the entire crypto market reached peak valuation in month of march, its facing marketing correction. But the dencun update is a positive step for Ethereum for staying relevant in rapidly changing markets.

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